

Revealing the role of hydrogen in reducing optical absorption and mechanical loss in magnetron-sputtered amorphous silicon for gravitational-wave detectors

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Hydrogenated *a*-Si films grown by magnetron sputtering have greatly reduced subgap optical absorption, down to 122 ppm (9.0 cm^{-1}) at 1064 nm, 6 ppm (0.4 cm^{-1}) at 1550 nm, and 2 ppm (0.1 cm^{-1}) at 2000 nm for quarter-wavelength-thick films, and they reduced room-temperature mechanical loss below 2×10^{-5} . The dangling bond density is reduced to 10^{17} cm^{-3} and the refraction index remains above that of crystalline silicon for all wavelengths. The above wavelengths are well below the Urbach edge of *a*-Si, and thus absorption is associated with deep-trap states. Although reductions in absorption and dissipation increase with increasing hydrogen content in the working gas, the hydrogen concentration in the films remains low (below 1 at.%), and neither absorption nor mechanical loss correlate with dangling bond density. Raman-derived bond angle deviation measurements of *a*-Si show a reduction in structural disorder as the films are further processed by vacuum annealing and post-hydrogenation, while *a*-Si:H films show a lower disorder in the as-deposited state without further reductions after thermal processing. These results suggest that the presence of hydrogen during film growth, acting as a catalyst, facilitates local rearrangement of atoms and removes strained Si-Si bonds that are more responsible for sub-band-gap absorption and dissipative mechanisms than dangling bonds.

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I. INTRODUCTION

Hydrogenated amorphous silicon (*a*-Si:H) has attracted considerable attention in optoelectronic, photonic, and photovoltaic devices and has been intensively studied for years [1–3]. Hydrogen is crucial to most uses of amorphous silicon, specifically for electronic devices such as photovoltaic and semiconductor *p*-*n* junctions, where the elimination of dangling bonds is essential to their performance. In the past, hydrogenation of amorphous silicon (*a*-Si) films was primarily investigated using chemical vapor deposition techniques, by either *in situ* hydrogenation (*in situ* H), or postgrowth hydrogenation in forming gas (post-H). *In situ* H incorporates hydrogen during growth, e.g., using silane as precursor gas, molecular H₂ or H ions, which produce *a*-Si:H of various H concentrations. Post-H relies on annealing in forming gas, a mixture of H₂ and N₂, or hydrogen plasma treatment [4]. For *a*-Si:H films deposited by chemical vapor deposition (CVD) techniques, specifically plasma-enhanced chemical vapor deposition (PECVD) and hot-wire chemical vapor deposition (HWCVD), it has been shown that deposition parameters, such as pressure, substrate temperature (*T*_s), and silane/H₂ ratio, can be used to produce films of desired electronic prop-

erties with H contents ranging from 1 to 20 at.% H [5–7]. CVD techniques, including PECVD, are generally unsuitable for optical coatings in interferometric metrology, such as those used in gravitational-wave (GW) detectors, due to limitations in achieving the required large-area uniformity and thickness control. As a result, PVD methods, particularly ion beam sputtering (IBS), are the standard in GW detector coatings, with magnetron sputtering (MS) serving as a closely related and experimentally accessible technique for exploring *a*-Si film properties in this context. Studies of PVD *a*-Si:H films have also demonstrated improved electronics properties, including increased optical band gap and minority carrier lifetime [8,9].

Although hydrogen is known to be essential in creating device quality *a*-Si, there is little to no correlation between the H content and the electronic properties, including both the optical band gap and the dangling bond density. Although most researchers believe that the primary role of H in these materials is to eliminate dangling bonds, there are orders of magnitude mismatch between the amount of H in the films and the eliminated dangling bonds, suggesting that the role of H is more complex. Previous work has shown that (i) the addition of H reduces structural disorder, and that (ii) disorder, rather than hydrogen content, is the fundamental determining factor in the optical band gap [10,11]. It is important to note that for most applications, though not all, increased optical absorption is desirable, particularly in photovoltaics, where absorption studies in *a*-Si:H are largely driven by the needs of the photovoltaic and semiconductor industries.

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Hydrogen exists in various forms within *a*-Si films. Nuclear magnetic resonance spectroscopy revealed that the local hydrogen environment in HWCVD *a*-Si:H comprises primarily bonded Si-H and secondarily molecular hydrogen (H₂), both of which are found in both clustered and isolated states [12]. In RF sputtered *a*-Si:H films, Tour *et al.* identified isolated Si-H responsible for the midgap infrared (IR) absorption band at 4990 nm, and Si-H₂ groups absorbing around 4794 nm [13]. They found that the Si-H₂ groups are located within voids present in the amorphous network. Hydrogen redistribution within the Si network starts above 100 °C [7], while the formation of H₂ takes place at around 250 °C [14], and the desorption of hydrogen out of the film above 350 °C [15].

The prevailing view attributes the better performance of *a*-Si:H to the removal of sub-band-gap states through the passivation of dangling bonds (DBs) via hydrogenation [16]. However, simulation work has shown that in addition to DB passivation, hydrogen is involved in the structural reorganization of *a*-Si by lowering the energy barrier required for the network relaxation process [17–19], consistent with Cody's discussion in Ref. [10]. Sriraman *et al.* proposed that H insertion into strained Si-Si bonds results in chemical annealing of *a*-Si, a technique developed in the early 1990s [20,21] that uses H radicals to eliminate these strained bonds via H-induced atomic rearrangements [17,18]. After such events, hydrogen atoms either stay bonded to these Si atoms, or leave and diffuse through the network [19]. Other literature from the 1980s and 1990s established the general principle that hydrogen promotes ordering in *a*-Si:H prepared by both CVD and PVD (see, e.g., [22–25]). Of these papers, only Ref. [22] measured the effect on optical absorption of sputtered films above 4500 nm (below 0.28 eV), far from the IR range of interest to gravitational wave detectors (from about 1000 to 2000 nm). Furthermore, all these results were on *a*-Si:H containing large amounts of hydrogen (typically ≥ 10 at.% H), which is known to be deleterious to the high index of refraction needed for GW detectors, and the effect on room-temperature mechanical loss, critical to low thermal noise in GW detectors, was not studied.

Amorphous silicon has shown potential to be used as a high-index reflective layer in dielectric mirror coatings used in GW detectors [26–37]. It has been studied by various PVD techniques, such as electron beam (e-beam) evaporation [26], MS [27], IBS [28,29] and reactive low-voltage ion plating (RLVIP) [30]. Its high refractive index [26,30,31], and low mechanical loss at cryogenic temperatures [32] and room temperature [28], which is directly proportional to the coating thermal noise in GW detectors, has made it a strong candidate to be employed in the next upgrade of current GW detectors, Advanced LIGO [33], and future detectors like LIGO Voyager [34], Cosmic Explorer [35] and Einstein Telescope [36]. Despite its excellent mechanical loss performance and consequently low thermal noise, *a*-Si has a significantly higher optical absorption than the currently used Ti-doped tantala (*a*-Ti:Ta₂O₅) layer at the operating laser wavelength of 1064 nm [33]. To mitigate the impact of the high absorption at 1064 nm in *a*-Si layers on the total coating absorption, a multilayer design has been proposed by Steinlechner *et al.* [37], in which *a*-Si is embedded deeper within the coating

stack. This design enhances reflectivity, while only a small fraction (0.35%) of the beam reaches the stack containing *a*-Si, thereby reducing its influence on the overall coating absorption, while still taking advantage of its low thermal noise.

Optical absorption in *a*-Si contains three distinct regions, well described in several papers by Cody *et al.* [10,38,39]. Above approximately 3.4 eV, absorption is very high in *a*-Si and virtually identical to that in crystalline Si (*c*-Si), unsurprisingly given their similar local structure and consequently similar conduction and valence bands. Below 3.4 eV, in what is sometimes called region C, absorption is an algebraic function of photon energy, with values of 10^3 – 10^5 cm⁻¹, governed by the excitation of bonded valence electrons to free-carrier conduction states, and hence dominated by the carrier densities of states in each band which are parabolic, as measured by Jackson *et al.* [40]. This region, when fit with an algebraic function, yields what is called the optical absorption gap E_g , sometimes called a Tauc or Cody gap depending on details of the fit, which varies between 1.6 and 1.9 eV for *a*-Si, depending on sample preparation. Notably, there is no sharp feature in the *a*-Si absorption spectrum near 1.6, 1.9, or 3.4 eV. Below ~ 2 eV, in region B, the absorption is an exponential function of energy and drops by orders of magnitude; this region is known as the Urbach tail, with a width known as the Urbach edge parameter E_0 that depends on the structural disorder induced by measurement temperature and sample preparation. The Urbach tail is governed by transitions from band tail states, which are broadened by disorder, to the conduction band. Finally, in region A at energies below ~ 1.3 eV, hence the wavelengths relevant to gravitational wave detectors, the absorption coefficient, which Cody *et al.* showed to be near 1 cm⁻¹ [39], becomes again algebraic and is dominated by excitations from deep-trap subband states, which depend on preparation conditions, to the conduction band. Region A, the region relevant to gravitational-wave detectors, may show impurity-induced peaks and absorption being nonmonotonic in energy. The work by Cody *et al.* focused on *a*-Si:H films grown by glow-discharge chemical vapor deposition, a technique that produces films unsuitable for interferometric applications due to their high surface roughness. In addition, these films contained significant amounts of hydrogen (~ 16 at.% H), which is known to substantially reduce the refractive index in *a*-Si:H [41].

The exponential Urbach tail is a characteristic feature of amorphous solids [42] and is also seen in crystalline materials, e.g., near the indirect gap of *c*-Si. Its breadth depends on sample preparation [10,43]. Pierce and Spicer discussed tail states in *a*-Si, pointing out that the sharpness of band edges is influenced by sample preparation, and the formation of tail states is more likely related to structural imperfections, such as voids, rather than the lack of long-range order [43]. Seager *et al.* reported different responses in spin density and absorption spectrum of polycrystalline Si and plastically deformed Si after post-hydrogenation and thermal annealing and excluded dangling bonds as sources for the Urbach tail in *a*-Si [44]. Cody *et al.* found that in *a*-Si:H, the width of the exponential tail (E_0) is determined by the combined effects of structural and thermally induced disorder in the amorphous network, and that E_g and E_0 are inversely related, specifically that increased

disorder in a -Si:H, whether due to increasing measurement temperature or structural disorder induced by specific preparation conditions, causes E_0 to increase and E_g to decrease [10]. Atomistic modeling of a -Si suggests that the tail states are primarily localized on stretched Si-Si bonds [45–47].

In contrast, much less is known about region A and the subgap states that lead to absorption below ~ 1.3 eV. This regime notably includes 1064 nm, the current wavelength used for gravitational-wave detectors, as well as the wavelengths contemplated for future detectors (1550 and 2000 nm). In Cody's work, it is notable that the optical absorption of a -Si:H is significantly lower than that of c -Si between 1.6 and 1.0 eV, or equivalently between 775 and 1240 nm, associated with the onset of appreciable absorption at the indirect gap of c -Si, which also shows an Urbach edge [38]. These subgap states may include dangling bond states, but are likely also created by impurities and other structural defects.

Previously, Molina-Ruiz *et al.* demonstrated that the post-hydrogenation of e-beam-grown a -Si 500-nm-thick films grown at room temperature reduces the absorption at 2000 nm to below 10 ppm (0.2 cm^{-1}), which, combined with a low-index material, meets the sub-ppm requirement for cryogenic GW mirror coating at this wavelength [26]. The same work also showed that post-H has a limited effect in the mechanical loss at room temperature, demonstrating that there are separate structural mechanisms involved in photon subgap absorption and mechanical energy dissipation. Mechanical loss at low temperature (< 100 K) has been shown to be reduced by either PVD growth at elevated temperatures [26] or by added hydrogen [48], due predominantly to structural relaxation.

In this paper, we present our findings on a -Si and a -Si:H films prepared using magnetron sputtering and hydrogenated either during or after deposition. In addition, postgrowth annealing in vacuum and post-hydrogenation in forming gas were performed on the films. We studied the optical absorption of MS a -Si:H in all these states at three infrared wavelengths: 1064 nm, which is used in the current LIGO detectors, and 1550 and 2000 nm, which have been proposed for the designs of next-generation GW detectors. Mechanical loss at room temperature was also measured, as well as dangling bond density, bond angle deviation, local strain, refractive index, and hydrogen concentration in the films. We observed a significant decrease in optical absorption and mechanical loss at room temperature, with only a small reduction in refractive index in both *in situ* hydrogenated and post-hydrogenated MS a -Si:H samples, which makes a -Si:H a viable high-index material candidate for future gravitational-wave detectors at all three wavelengths. Our work reveals that this improvement in optical and mechanical properties is not due to passivation of dangling bonds. Instead, we propose that weak Si-Si bonds are primarily responsible for sub-band-gap (below 1.2 eV) absorption in a -Si and dissipation of mechanical energy. Hydrogen acts as a catalyst that promotes local rearrangement of atoms and eliminates weak bonds that cause absorption and dissipation. The films in this study have less than 1 at.% H, unlike previous CVD and PVD experiments. It is likely that some of the molecular H_2 in the $\text{Ar}:\text{H}_2$ gas mixture used for this work is dissociated into atomic H, which would enhance its effectiveness. Our results are consistent with the literature on chemical annealing of a -Si:H, but extend it by showing

reduced mechanical loss and optical absorption in the energy range relevant to GW detectors, without significant reduction in refractive index. This study provides insights into the mechanisms of photon absorption and energy dissipation in a -Si:H and offers a path to suppress them, allowing the development of high-performance a -Si:H dielectric mirror coatings for gravitational-wave detectors and other devices.

II. EXPERIMENTAL DETAILS

A. Sample preparation

Amorphous silicon films were grown by magnetron sputtering using 99.999% Ar working gas with different H_2 mixtures, containing between 0% and 20% H_2 . An undoped Si target (99.999% purity) was used with a dc pulsed power supply and a substrate-to-target distance of 15.2 cm. Pure Ar was introduced from the sputtering chamber's bottom, while an 80% Ar – 20% H_2 gas mixture was injected through a feedthrough positioned near the substrate plate. The flow rates of pure Ar and 80% Ar – 20% H_2 gas were separately tuned to obtain a total combined flow rate that ranged between 10 and 14 sccm depending on the hydrogen concentration in the working gas. For simplicity, we will refer to the hydrogen content present in the working gas, i.e., 0% H_2 , 5% H_2 , 10% H_2 , 15% H_2 , and 20% H_2 . The chamber gas pressure was maintained at 1.5 mTorr in all cases. The base pressure was $< 2 \times 10^{-8}$ Torr. Films were grown at $0.5 \text{ \AA/s}$ with thicknesses ranging between 305 and 378 nm. The substrate temperature T_s was chosen to be nominally room temperature RT ($\sim 60^\circ\text{C}$) or 360°C .

Postgrowth treatments included annealing in vacuum or in 99.9999% forming gas (5% H_2 in N_2). Vacuum annealing was performed in the same chamber where the films were grown at 360°C for 3 h (with no added gas). Post-H was performed in a standard horizontal quartz tube furnace using high-purity forming gas at 400°C for 3 h, with a flow rate of 20 mL/min. Hydrogen diffusion within a -Si:H during thermal annealing is known to occur at moderate temperatures, with the onset reported as low as 155°C [49], and activation energies typically in the range of 1.2–1.5 eV, consistent with trap-limited and dispersive diffusion mechanisms [50,51]. These processes become increasingly efficient at elevated temperatures, such as the 360°C used in our study, enabling hydrogen to passivate dangling bonds and facilitate structural relaxation [26].

B. Characterization methods

Ion beam analysis was performed in a NEC model 5SDH Pelletron tandem accelerator, with a beam energy of α particles with 3050 keV. In this tool, we performed Rutherford backscattering spectrometry (RBS) to obtain the composition and areal density of the films, and elastic recoil detection analysis (ERDA) to quantify their hydrogen content and depth profile. The films' atomic density is calculated as the ratio between the areal density of the films and their thickness, obtained by using a KLA Tencor ASIQ profilometer.

The density of dangling bonds ρ_{DB} was measured using electron paramagnetic resonance (EPR) spectroscopy. The measurement was performed with a Bruker ELEXSYS E580 EPR spectrometer using an X-band ER 4123D CW-Resonator operating at 9.62 GHz. A microwave power of 1.5 mW and a

magnetic-field modulation amplitude of 2 G were used to obtain spectra between 3375 and 3475 G. The contribution from the substrates used for these measurements, high-resistivity silicon, was determined using a bare substrate, and a KCl calibration standard was used to determine ρ_{DB} of the films.

Raman spectroscopy was performed using an inVia Renishaw micro-Raman/PL system equipped with a 488 nm laser. The main feature of the *a*-Si Raman spectrum is the transverse optic (TO) peak, which indicates changes in the short-range order of the *a*-Si network. Beeman *et al.* showed that the increase in the width of the TO peak at 480 cm^{-1} in *a*-Si, Γ , is due to an increase in the deviation of the bond angle between Si atoms, $\Delta\theta$, with respect to the tetrahedral angle in *c*-Si: $\Delta\theta = (\Gamma - 15)/6$ [52]. Strubbe *et al.* established that the shift of the TO peak in *a*-Si:H correlates with a change in local bond strain: $\text{tr}(\varepsilon) = (\omega - \omega_0)/s$, where ω is the TO peak position measured in cm^{-1} , $\omega_0 = 480\text{ cm}^{-1}$ is the TO peak position in fully relaxed *a*-Si, and $s = 460 \pm 10\text{ cm}^{-1}$ is a fitting parameter [53]. Positive values of $\text{tr}(\varepsilon)$ correspond to a tensile local strain, meaning Si atoms that are farther apart than in *c*-Si, while negative values correspond to a compressive local strain, meaning Si atoms that are closer together than in *c*-Si. We note that macroscopic tensile (or compressive) strain, measured typically by substrate curvature, refers to Si-Si bonds out of plane being shorter (or longer) than in plane, which can occur with either tensile or compressive local strains (as discussed in Ref. [54]).

The refractive index n of the films was measured using a Horiba UVISSEL ellipsometer in a spectral range from 200 to 2100 nm at an angle of incidence of 60° and in combination with DeltaPsi2 software to adjust the optical parameters with a Tauc-Lorentz model.

Optical absorption at 1064, 1550, and 2000 nm was measured using photothermal common-path intergerometry (PCI) [55]. This technique uses a pump-probe method to measure disturbance in the probe beam caused by a local temperature increase in an optic, which arises from absorbed light power from the pump beam. The signal is compared to a calibration sample to determine absorption strength. In this work, the optical absorption is calculated considering the interference effect due to the refractive index difference between film and substrate [26], after which it is normalized to the thickness t of a quarter-wavelength (QWL) thick film ($t = \lambda/4n$, where λ is the laser wavelength and n is the refractive index of the film). The optical absorption reported in this work has an error of $\pm 7\%$ of the measured values and includes all alignment and calibration uncertainties. The films were grown on Corning 7980 and 7979 fused silica 1-in.-diam and 1/4-in.-thick fused silica substrates.

The mechanical loss of *a*-Si and *a*-Si:H films was measured at room temperature using the gentle nodal suspension (GeNS) technique, which is described in detail in the work of Cesarini *et al.* [56]. For this measurement, films were deposited onto 2-in.-diam and 1/2-mm-thick fused silica wafers.

III. RESULTS

A. Hydrogen concentration and depth profile

Hydrogen content and its depth profile are obtained from ERDA measurements for all films presented in this paper. By

TABLE I. Atomic concentration of hydrogen at.% H in MS *a*-Si and *a*-Si:H films grown at room temperature (RT) or 360°C and at 1.5 mTorr using different amounts of H_2 in the working gas.

Sample state	Growth conditions	at.% H
as-deposited	RT, 0% H_2	^a
	RT, 5% H_2	0.29 ± 0.04
	RT, 10% H_2	0.35 ± 0.04
	RT, 15% H_2	0.97 ± 0.10
	RT, 20% H_2	1.16 ± 0.12
	360°C , 0% H_2	^a
	360°C , 5% H_2	0.07 ± 0.04
	360°C , 10% H_2	0.14 ± 0.03
	RT, 0% H_2	^a
vacuum annealed	RT, 5% H_2	0.28 ± 0.03
	RT, 10% H_2	0.36 ± 0.04
	RT, 15% H_2	0.95 ± 0.10
	RT, 20% H_2	1.18 ± 0.12
	360°C , 0% H_2	^a
	360°C , 5% H_2	0.08 ± 0.04
	360°C , 10% H_2	0.12 ± 0.03
	RT, 0% H_2	0.04 ± 0.11 ^b
	RT, 5% H_2	0.28 ± 0.03
post-hydrogenated	RT, 10% H_2	0.35 ± 0.04
	RT, 15% H_2	0.97 ± 0.10
	RT, 20% H_2	1.16 ± 0.12
	360°C , 0% H_2	0.03 ± 0.21 ^b
	360°C , 5% H_2	0.08 ± 0.04
	360°C , 10% H_2	0.14 ± 0.02

Systematic uncertainties are estimated at $\pm 10\%$ of the measured values, while total errors combine statistical and systematic errors in quadrature.

^aThe measurement result is below the detection threshold.

^bHydrogen is not continuously distributed through the film [57].

comparing the *in situ* H samples, and samples deposited with 0% H_2 and then post-H, in Table I we observe that the *in situ* H process is more effective at introducing hydrogen into *a*-Si. The RT-deposited *in situ* H film with 20% H_2 has the highest H content at 1.16 at.% H. The at.% H of films grown with 0% H_2 , in the as-deposited and vacuum-annealed states, is below the 0.01 at.% H detection threshold of the ERDA technique. After post-hydrogenation, hydrogen is found primarily near the surface and at the interface between film and substrate [57]. Films grown with *in situ* H show that hydrogen is more evenly distributed, with a small and continuous reduction in at.%H between the surface and the interface. This hydrogen distribution becomes more homogeneous as films are grown with larger concentrations of H_2 in the working gas. Additional details on the hydrogen content and depth profiles of the films are described in the Supplemental Material [57].

We note that the amount of hydrogen incorporated in MS *a*-Si:H films is much lower than that found in other studies on *in situ* H *a*-Si:H, including RF sputtering (up to 16 at.%) [59] and HWCVD (up to 4 at.%) [60], and post-H *a*-Si:H, such as RF sputtering (up to ~ 8 at.%) [9] and e-beam (up to 1.45 at.%) [26]. We also note that oxygen was measured by RBS and a surface oxide layer was found in all films. However, no oxygen (above 0.1 at.%) was found in any of the films.

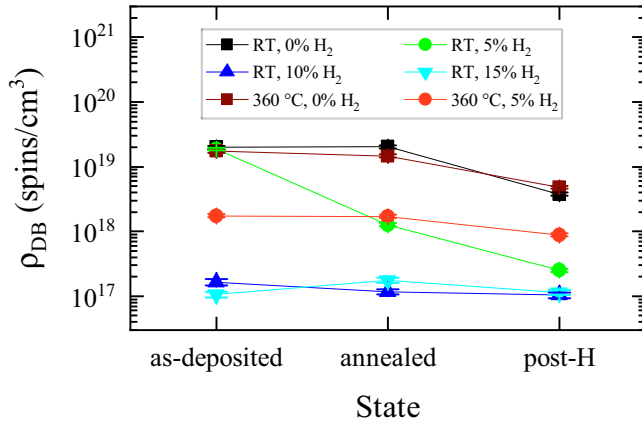


FIG. 1. Dangling bond density ρ_{DB} of a -Si deposited using 0% H_2 , and a -Si:H deposited using different concentrations of H_2 in the working gas, and at room temperature (RT) or 360 °C, as a function of the as-deposited, annealed, and post-H states. Lines are guides to the eye.

Previously, Molina-Ruiz *et al.* reported that in HWCVD a -Si:H films, the hydrogen content does not change on annealing at 300 °C in vacuum [61]. We similarly found that the hydrogen incorporated during the deposition process remains in the a -Si:H film, within the resolution of the technique of ± 0.01 at.% H, even after annealing at 360 °C for up to 6 h in high vacuum [57]. This indicates that high-temperature treatments, at least up to 360 °C, can be carried out on MS a -Si:H without losing hydrogen.

B. Dangling bond density

The unhydrogenated a -Si films, grown at either RT or 360 °C, have a dangling bond density ρ_{DB} of 2×10^{19} spins/cm³, a typical high level for a -Si, which remains unchanged by vacuum annealing (see Fig. 1). Films grown at RT with 5% H_2 do not show a significant change in the as-deposited ρ_{DB} , but after vacuum annealing of this film, there is an order of magnitude drop (to 10^{18} spins/cm³), which is then further reduced by post-H. Films grown at 360 °C in 5% H_2 have ρ_{DB} reduced by an order of magnitude, which is then little changed by vacuum annealing or post-H. Increasing the hydrogen content in the working gas to 10% and 15% further reduces ρ_{DB} by another order of magnitude (to 10^{17} spins/cm³), with no significant further reduction after vacuum annealing or post-H. These lowest values are still somewhat higher than those found in CVD-prepared *device quality* a -Si:H (on the order of 10^{16} spins/cm³), likely because the hydrogen content in the present films is quite low compared to that found in CVD films. We note that the 360 °C *in situ* H sample, grown with 5% H_2 , shows a significantly lower ρ_{DB} (1.8×10^{18} spins/cm³) than the RT-grown *in situ* 5% H_2 film, despite having a significantly lower H content than the RT-grown *in situ* H film (0.09 vs 0.26 at.% H, respectively), demonstrating again the commonly observed fact, for both PVD and CVD films, that while hydrogen is essential to reducing ρ_{DB} , a necessary condition for electrical properties of a -Si:H, the at.% H in the films does not directly correlate to ρ_{DB} .

After vacuum annealing, Fig. 1 shows that the hydrogenated film grown at RT with 5% H_2 exhibits a significant reduction (more than an order of magnitude) in ρ_{DB} , while ρ_{DB} of the other films barely changes. Post-H also reduces ρ_{DB} of the unhydrogenated samples to approximately one-fourth of their initial value. The a -Si:H films grown at RT with 10% H_2 and 15% H_2 in the as-deposited state show the lowest ρ_{DB} of 1×10^{17} spins/cm³, which does not change significantly after vacuum annealing and post-H. We note that these films never reached ρ_{DB} values of the order of 10^{16} spins/cm³ as seen in device-quality films, probably due to the limited H uptake.

In general, we have identified two effective means to remove dangling bonds of MS a -Si:H films: (1) *in situ* H, both at RT and 360 °C, and (2) post-H, with the lowest ρ_{DB} found for RT *in situ* H films grown with 10% and 15% H_2 . The effectiveness of dangling bond passivation in MS-grown a -Si:H depends strongly on the hydrogen concentration during growth, reflecting two distinct hydrogen incorporation mechanisms: (1) up to 5% H_2 , passivation requires postdeposition thermal activation and additional hydrogen incorporation, likely due to insufficient hydrogen availability during deposition to fully terminate dangling bonds; and (2) at 10% H_2 and above, the *in situ* hydrogen density in the films is high enough to enable direct passivation during growth, making additional annealing or post-hydrogenation steps unnecessary.

C. Bond angle deviation and local strain

Figure 2 shows the bond angle deviation $\Delta\theta$ of the a -Si and a -Si:H films grown at different substrate temperatures: RT (top panel) and 360 °C (bottom panel). In the case of films grown with 0% H_2 , $\Delta\theta$ significantly decreases after vacuum annealing and post-H, aligning with the findings in e-beam a -Si [26]. Films grown at RT using 5% to 20% H_2 show a lower $\Delta\theta$, between 8.5° and 9.1°, which does not change significantly with vacuum annealing or post-hydrogenation.

The local strain $\text{tr}(\epsilon)$ obtained for all films studied in this paper, and shown in Fig. 3, is positive, which indicates that the films are under tensile local strain, i.e., the Si-Si bonds are stretched on average relative to c -Si. The analysis of $\text{tr}(\epsilon)$ of these films shows a somewhat less straightforward effect of hydrogen in the Si network. In the as-deposited state, films grown using 5% H_2 are more locally strained than films grown with higher at.% H_2 concentrations, or even grown without hydrogen at all. However, in the annealed state, films grown with 15% and 20% H_2 increase their local strain significantly above that of the other films. After post-hydrogenation, the local strain of all films either drops or remains approximately the same. Films grown at RT and post-H show a local strain of $(1.6 \pm 0.3)\%$, whereas those grown at 360 °C show a smaller local strain of $(0.6 \pm 0.1)\%$, suggesting that elevated growth temperature helps to produce films with lower local strain.

D. Refractive index

The refractive index n of the films studied in this paper is shown in Fig. 4. As the wavelength increases, n of all samples decreases, which is expected since 2000 nm is farther

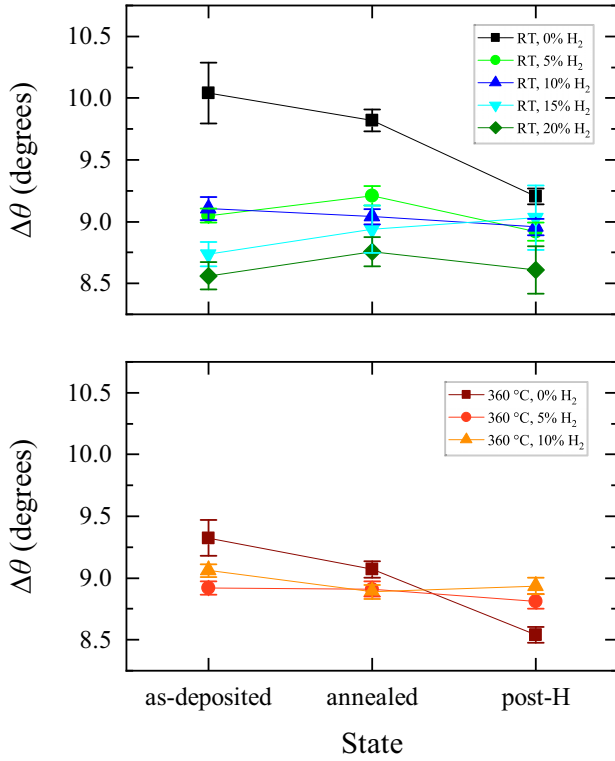


FIG. 2. Bond angle deviation $\Delta\theta$ of MS *a*-Si and *a*-Si:H deposited using different concentrations of H_2 in the working gas, and at RT (top) and at 360 °C (bottom) as a function of the as-deposited, annealed, and post-hydrogenated states. Lines are guides to the eye.

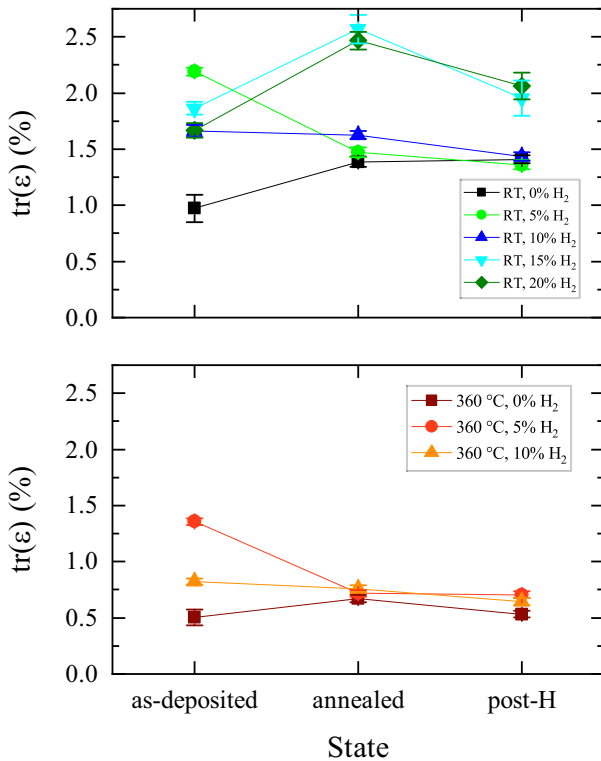


FIG. 3. Local bond strain $tr(\epsilon)$ of MS *a*-Si and *a*-Si:H deposited using different concentrations of H_2 in the working gas, and at RT (top) and at 360 °C (bottom) as a function of the as-deposited, annealed, and post-hydrogenated states. Lines are guides to the eye.

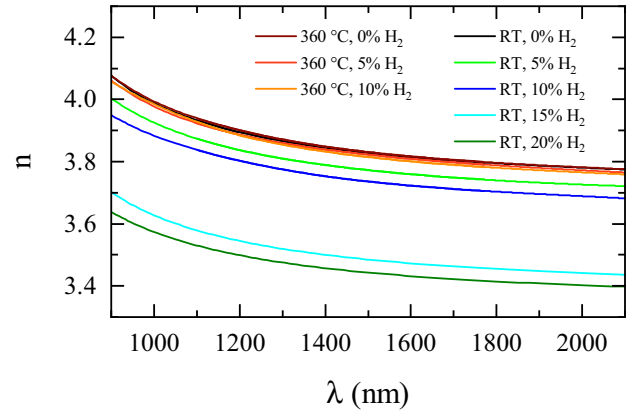


FIG. 4. Refractive index n as a function of wavelength λ for *a*-Si and *a*-Si:H films grown at RT and 360 °C and with different hydrogen content in the working gas. All films were measured in the post-H state. The three films grown at 360 °C (with 0%, 5%, and 10% H_2) have nearly identical $n(\lambda)$ that overlaps with that of the film grown at RT with 0% H_2 .

from the silicon band gap than 1064 nm. All films grown at RT show a monotonic reduction of n as the at.% H in the film increases. Nonetheless, n is larger than that of *c*-Si for all films, except for films grown with 20% H_2 , which have comparable values. The refractive indices of films grown at 360 °C are comparable to those of films grown at RT with 0% H_2 , regardless of the H_2 concentration used during growth (at least up to 10% H_2).

The reduction in n observed with the increase of the at.% H in the films grown at RT is not associated with a lower atomic density, since the atomic density of these films is comparable to that of the *a*-Si film grown at RT using 0% H_2 [57]. Instead, we attribute the changes to an increased band gap E_g and a reduced band tail density of states, hence reduced E_0 , both caused by structural relaxation changes induced by hydrogen.

E. Optical absorption

The optical absorption of MS *a*-Si and *a*-Si:H films measured at 1064, 1550, and 2000 nm, and normalized to the thickness of a QWL-thick film, as well as their absorption coefficient α , are shown in Table II. In as-deposited *a*-Si:H films, *in situ* H significantly reduces their optical absorption. Higher H_2 concentration in the working gas, from 5% to 10% H_2 , decreases absorption for both growth temperatures (an exception at 2000 nm will be discussed below), and from 5% to 15% H_2 for films grown at room temperature. Vacuum annealing is more efficient in reducing absorption in RT-deposited films than in those grown at elevated T_s , consistent with findings in e-beam *a*-Si [26]. Post-H treatment is equally effective in reducing absorption in initially unhydrogenated and hydrogenated samples during growth. The lowest absorption at 1064 and 1550 nm is observed for the *a*-Si:H film grown at RT with 10% H_2 and post-H, and the lowest absorption at 2000 nm is for the *a*-Si:H film grown at RT with 15% H_2 and vacuum annealed (values highlighted in bold in Table II). The performance at 1064 nm, yielding 122 ppm (9 cm^{-1}), at 1550 nm, yielding 6 ppm (0.4 cm^{-1}), and at 2000 nm,

TABLE II. Optical absorption in ppm for QWL-thick MS *a*-Si and *a*-Si:H films (with the corresponding absorption coefficient α in cm^{-1} shown in parentheses). Films are measured using laser wavelengths of 1064, 1550, and 2000 nm in the as-deposited, vacuum annealed (at 10^{-8} Torr at 360°C for 3 h) and post-H states (at atmospheric pressure in high purity forming gas at 400°C for 3 h). The QWL thickness t of the films is calculated using $t = \lambda/4n$, where λ is the laser wavelength and n is the film refractive index reported in Fig. 4. The absorption coefficient is defined as $\alpha = -\ln(1 - A)/t$, where A is the optical absorption in ppm and t is the film thickness in cm. Deposition conditions indicate the growth temperature of the films: RT or 360°C , hydrogen concentration in the working gas used during growth: 0%, 5%, 10%, 15%, or 20% H_2 , and measured thickness. The absolute error for both optical absorption and absorption coefficient is $\pm 7\%$ of the reported value. Values in bold highlight the lowest absorption obtained at each wavelength. The absence of a value indicates that the film is not measured for that wavelength and state.

Deposition conditions	as-deposited			vacuum annealed			post-hydrogenated		
	1064	1550	2000	1064	1550	2000	1064	1550	2000
RT, 0% H_2 , 364 nm		11081 (1100)	2089 (158)		1020 (101)	183 (14)	2413 (359)	422 (42)	47 (3.6)
RT, 5% H_2 , 331 nm ^a		2143 (208)	1822 (136)	1089 (161)	1455 (142)	91 (6.8)	594 (88)	68 (6.6)	21 (1.6)
RT, 10% H_2 , 328 nm	2909 (424)	498 (72)	241 (35)	266 (26)	19 (1.8)	39 (3.7)	122 (9.0)	5.8 (0.42)	14 (1.0)
RT, 15% H_2 , 312 nm	5976 (817)	262 (24)	319 (22)	94 (13)	7.6 (0.68)	1.9 (0.13)	204 (28)	12 (1.1)	12 (0.84)
RT, 20% H_2 , 305 nm	3787 (508)	155 (14)	122 (8.3)	362 (48)	82 (7.3)	102 (7.0)	749 (100)	74 (6.6)	188 (13)
360 $^\circ\text{C}$, 0% H_2 , 378 nm		2349 (232)	212 (16)		2108 (209)	178 (14)	6302 (939)	968 (96)	69 (5.2)
360 $^\circ\text{C}$, 5% H_2 , 337 nm	1041 (156)	243 (24)	121 (9.1)	900 (135)	228 (23)	176 (13)	778 (117)	82 (8.1)	71 (5.3)
360 $^\circ\text{C}$, 10% H_2 , 331 nm	374 (56)	68 (6.7)	141 (11)	433 (65)	49 (4.8)	127 (9.6)	324 (49)	25 (2.5)	50 (3.8)

^aAdditional annealing after the post-hydrogenated state, for a total of 27 h of vacuum annealing and 3 h of post-H, reduced the film's absorption only at 1064 nm, from 594 ppm (88 cm^{-1}) to 353 ppm (52 cm^{-1}).

yielding 1 ppm (0.1 cm^{-1}), surpasses that of the e-beam RT-deposited *a*-Si after post-hydrogenation, which yielded 504, 97, and 3 ppm, respectively, at each wavelength [26]. The observation that some MS *a*-Si:H films have higher absorption at 2000 nm than at 1550 nm indicates that the absorption is related to an impurity peak or other nonexponential band-gap states.

The optical absorption of films grown at RT with 20% H_2 is higher for the vacuum-annealed and post-H states than that of films grown at RT with 10% or 15% H_2 . However, in the as-deposited state, the absorption of the *a*-Si:H film grown at RT with 20% H_2 is lower at 1550 and 2000 nm, and not clearly worse at 1064 nm.

The overall reduction in absorption due to hydrogenation in MS *a*-Si films is similar to that achieved by e-beam films, e.g., both e-beam and MS RT films measured at 1550 nm show a twofold reduction in absorption after post-H [26]. The larger reduction in the absorption of MS films compared to that of e-beam films is achieved because of *in situ* H. Table II shows the absorption values of MS *a*-Si:H films grown at RT with 10% and 15% H_2 in the post-H and annealed states, respectively, which are orders of magnitude lower at all measured wavelengths than that of MS *a*-Si films grown at RT with 0% H_2 in the as-deposited state.

Additional vacuum annealing, up to 30 h, of films deposited at RT with 0% and 5% H_2 that have undergone post-H resulted in a decrease in absorption only at 1064 nm, while the absorption at 1550 and 2000 nm remained unchanged,

suggesting that different trap states are involved in the optical absorption at 1064, 1550, and 2000 nm. The RT *in situ* H with 5% H_2 *a*-Si:H film, after a total of 27 hours of vacuum annealing and 3 hours of post-H, achieved an absorption of 353 ppm at 1064 nm, for a 68 nm QWL-thick layer, compared to 594 ppm obtained after 3 h of annealing and 3 h of post-H. This is a 40% reduction of the optical absorption at 1064 nm by increasing the annealing time fivefold.

The films studied in this work were sequentially measured in the as-deposited, vacuum-annealed, and post-hydrogenated states, and we do not know whether the intermediate step of vacuum annealing is needed to achieve the absorption values reported after post-H. However, it was shown that the intermediate annealing step is not necessary in *a*-Si [26].

F. Mechanical loss

The mechanical loss at room temperature Q_{RT}^{-1} of MS *a*-Si and *a*-Si:H films is shown in Fig. 5. There is a moderate reduction of Q_{RT}^{-1} with subsequent treatments for films grown at RT, while films grown at 360°C do not show any significant change. As an overall trend, a higher hydrogen concentration during growth produces films with lower Q_{RT}^{-1} . The *in situ* H *a*-Si:H film, grown at RT with 15% H_2 in the vacuum-annealed and post-hydrogenation states, shows a Q_{RT}^{-1} of 1.8×10^{-5} , the lowest mechanical loss measured at room temperature to date for amorphous silicon, with or without hydrogen.

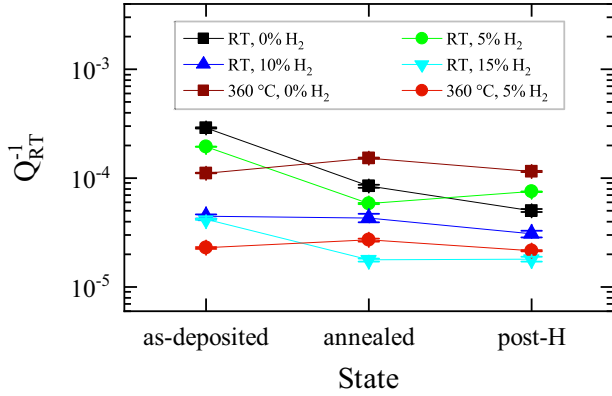


FIG. 5. Mechanical loss measured at room temperature Q_{RT}^{-1} of MS *a*-Si and *a*-Si:H films as a function of their state. Films were grown at RT and 360 °C using different concentrations of H₂ in the working gas. Lines are guides to the eye.

IV. DISCUSSION

The presence of hydrogen during growth (*in situ* H) leads to lower absorption in all states (as-deposited, vacuum-annealed, and post-hydrogenated) compared to unhydrogenated films, for the three wavelengths studied, as shown in Table II. We remind the reader that these wavelengths are well below the band gap and the Urbach edge of *a*-Si, with or without H, and hence the absorption is associated with deep-trap states. This region is much less studied in the literature as it is not relevant to either electronic properties or photovoltaic processes. Increasing the H₂ concentration in the working gas leads to lower absorption. The lowest absorption at 1064 and 1550 nm is observed for films grown at RT using 10% H₂ in the post-hydrogenated state, and at 2000 nm for films grown at RT using 15% H₂ in the vacuum-annealed state. Increasing the hydrogen concentration beyond these levels during RT growth results in increased absorption, indicating that there is an optimal hydrogen concentration that minimizes the absorption of MS *a*-Si:H films.

The lowest absorption for QWL-thick films at 1064 nm is 122 ppm (9.0 cm⁻¹), at 1550 is 5.8 ppm (0.42 cm⁻¹), and at 2000 nm is 1.9 ppm (0.13 cm⁻¹). The absorption at 1550 nm of 6 ppm is significantly lower than previously reported values for IBS *a*-Si (913 ppm) [31] and RLVIP *a*-Si (373 ppm) [30], all normalized to 106-nm-thick QWL film. Although the absorption at 1064 nm (122 ppm) still exceeds the desired ppm-level absorption, it represents a significant improvement compared to IBS *a*-Si (3396 ppm) [31], which opens up possibilities for applying the design of multimerial coatings proposed by Steinlechner *et al.* to current GW detectors [37].

The mechanical loss measured at room temperature Q_{RT}^{-1} shows that *in situ* H leads to lower loss compared to unhydrogenated films at both growth temperatures, RT and 360 °C, and for all at.% H concentrations (see Fig. 5). This noticeable reduction in Q_{RT}^{-1} is achieved by the film being exposed to hydrogen during growth, despite the very low amount of hydrogen that is incorporated into the film (less than 1 at.% H). In addition, annealing is effective in reducing Q_{RT}^{-1} only for films deposited at RT.

Dangling bonds are known to create trap states deep in the band gap, acting as recombination centers for charge carriers, and thus degrading the electrical properties of silicon. However, they are clearly not the dominant source of optical absorption in the low-energy region A (below ~1.2 eV), which arises from transitions between deep-trap and band states. Recently, Molina-Ruiz *et al.* showed that there is no correlation between the dangling bond density and the mechanical loss of *a*-Si [62]. In this work, we do not observe any correlation either between the dangling bond density and the mechanical loss at room temperature; three films with the same ρ_{DB} in the as-deposited state [RT, 0% H₂ (black square), 360 °C, 0% H₂ (brown square), and RT, 5% H₂ (green circle) in Fig. 1], show very different Q_{RT}^{-1} (same color code in Fig. 5). Most notably, there is no correlation between dangling bond density and optical absorption, e.g., both RT films grown with 0% and 5% H₂ have the same ρ_{DB} but an order of magnitude difference in optical absorption across all three wavelengths. Additionally, the film grown at RT with 5% H₂ shows a 20-fold reduction in ρ_{DB} on annealing and a further fivefold reduction after post-H processing, while its optical absorption decreases far less after annealing (factors from 30% to 100% depending on the measurement wavelength), followed by a fourfold reduction after post-H.

Our results align with those reported by Seager *et al.*, where a reduction in the sub-band-gap optical absorption (region A) of crystalline and polycrystalline silicon, including the three IR wavelengths measured in this paper, is observed without any measurable changes in ρ_{DB} [44]. MS *a*-Si:H films exhibit absorption levels as low as a few parts per million for QWL thickness, lower than that reported in e-beam *a*-Si:H [26]. We have found that growing or annealing at high substrate temperatures alone (without hydrogen) is sufficient to induce a significant reduction in both Q_{RT}^{-1} and infrared absorption of *a*-Si, with no significant change in ρ_{DB} . These results highlight that unpaired electrons, i.e., dangling bonds, are not the dominant factor determining mechanical loss at room temperature or infrared absorption in *a*-Si, despite being critical to electronic properties such as electron-hole recombination in photovoltaic cells.

Similarly to dangling bond density, neither bond angle deviation nor local strain are correlated with either mechanical loss at RT or IR optical absorption. As previously seen in *a*-Si [26], the bond angle deviation of MS *a*-Si films decreases with either heat provision (higher growth temperature and vacuum annealing), hydrogen introduction (*in situ* H), or a combination of both (post-H). As shown in Fig. 2, *in situ* hydrogenation is the most effective in reducing $\Delta\theta$ to levels where additional treatment does not significantly further reduce it. The *in situ* H *a*-Si:H films grown at RT in the as-deposited state exhibit lower $\Delta\theta$ as their at.% H increases, although their differences reduce after post-growth treatments. These results explain the lower optical absorption seen in *in situ* H films in the as-deposited state, and their better performance than *a*-Si deposited with 0% H₂ after vacuum annealing and post-H (see Table II).

These results are explained by considering that hydrogen plays a greater role than merely passivating dangling bonds, actively promoting structural relaxation during growth [10,23–25]. Given that molecular hydrogen in the Ar:H₂ plasma is likely dissociated into atomic and ionic species

during growth by MS, our process resembles what has been described as chemical annealing [20,21]. Sriraman *et al.* proposed several mechanisms of chemical annealing in *a*-Si:H, identified through molecular dynamics simulations, by which hydrogen is inserted into weak Si–Si bonds and assists in the relaxation, breakage, and formation of bonds, allowing the transition to a more energetically stable configuration [17,18]. The similar $\Delta\theta$ observed in Fig. 2 of the four as-deposited *in situ* H films (5–20% H₂) compared to the one grown without hydrogen (0% H₂), despite their quite different hydrogen content (from 0.29 to 1.16 at.% H shown in Table I), implies that the effectiveness of hydrogenation in promoting structural relaxation is not proportional to the amount of hydrogen in the film. This suggests that hydrogen acts as a catalyst, which reduces energy barriers and enables atomic rearrangement. Consequently, the resulting structure exhibits a higher degree of order (lower $\Delta\theta$) compared to unhydrogenated *a*-Si. The resulting structural change is likely to be permanent (because it is a lower-energy state), and thus it should not be susceptible to hydrogen leaving the film in later processing stages, which might be required for the fabrication of GW mirror coatings. The absence of further improvement with subsequent processing steps (vacuum annealing and post-hydrogenation) suggests that these *a*-Si:H films have already reached the optimal achievable structural configuration for these growth conditions. We suggest that the use of (i) hydrogen concentrations in the working gas between 10 and 15%, and (ii) growth temperatures slightly above room temperature, could lead to further reductions in the optical absorption and mechanical loss of MS *a*-Si:H films.

Hydrogen-induced structural relaxation is thus found to play a crucial role in suppressing sub-band-gap (region A) absorption (shown in Table II). The results presented in this work suggest that exposure to a hydrogen-rich environment during growth assists in the gradual removal of deep-trap states over time, even without a significant improvement in structural disorder measured by $\Delta\theta$. Therefore, defects that do not affect $\Delta\theta$, and are distinct from dangling bonds, which show no correlation with optical absorption at infrared wavelengths, create electronic states that contribute to IR (below 1.3 eV) photon absorption in *a*-Si:H. These defects are likely charged point defects, similar to those observed in *c*-Si, where charged divacancies, self-interstitials, and impurities introduce states in the band gap [63–65].

The higher absorption at 2000 nm compared to 1550 nm observed in most *in situ* H films grown with 10% or higher H₂ concentration, when the film absorption is below 100 ppm (see Table II), is not explained by conventional wisdom about absorption and band tails and is a clear sign of deep-trap or localized states from which carriers are excited into the conduction band by absorbing light [66,67]. It is likely that some particular defects in our MS *a*-Si:H films are responsible for this effect. Obtaining a full optical spectrum in the infrared and near-visible wavelength range would help to precisely identify these defects and their contributions to the optical absorption in *a*-Si:H.

Hydrogen-induced structural relaxation also helps to reduce mechanical loss at room temperature, as observed in Fig. 5. However, vacuum annealing and post-hydrogenation are not as effective in reducing Q_{RT}^{-1} of *in situ* H films as they

are for optical absorption. We hypothesize that the primary role of annealing and post-H in films with *in situ* H is to assist in the local rearrangement of atoms, thereby eliminating weak bonds that introduce localized electronic states. The resulting reduction in localized states, and thus in sub-band-gap absorption, continues over time, as heat enhances atomic mobility, until a saturation point is reached for a given film. In contrast, suppressing mechanical loss from energy dissipation mechanisms requires an overall strong and rigid bonding network. Achieving this requires not only local atomic rearrangement, but also reduction of other defects, such as divacancies and self-interstitials, which are difficult to detect directly and are likely not modified by post-H [based on the observation that $\text{tr}(\epsilon)$ does not significantly change].

The understanding of the role of hydrogen derived from this work could help reduce optical absorption and mechanical loss in other Si-based materials for the development of mirror coatings of GW detectors, such as SiN_x and SiO_xN_y.

V. CONCLUSIONS

Hydrogen plays a crucial role in reducing optical absorption and room-temperature mechanical loss of *a*-Si at infrared wavelengths. Our study reveals that magnetron-sputtered hydrogenated amorphous silicon films, grown at room temperature with H₂ concentrations of 10% and 15% in the working gas, exhibit extremely low-sub-band gap absorption at all three infrared wavelengths of interest to gravitational-wave detector designs. Hydrogenated amorphous silicon, grown at room temperature using 15% H₂ in the working gas, at 2000 nm has a refractive index of 3.44 and an optical absorption of 1.9 ppm for a quarter-wavelength-thick film, or equivalently an absorption coefficient of 0.13 cm^{−1}, and a mechanical loss measured at room temperature of 1.8×10^{-5} , the lowest room-temperature mechanical loss measured to date in amorphous silicon, both with and without hydrogen. As previously observed in *a*-Si [26,30], it is reasonable to expect that the mechanical loss of *a*-Si:H films will be lower at reduced temperatures, such as those planned for cryogenic gravitational-wave detectors.

Structural analysis (Raman spectroscopy and electron paramagnetic resonance) shows that hydrogen plays a dual role in amorphous silicon; in addition to the passivation of dangling bonds, it facilitates the rearrangement of Si–Si bonds, resulting in reduced atomic disorder, as measured by bond angle deviation. Hydrogen likely acts as a catalyst that promotes relaxation and rearrangement of weak Si–Si bonds, and then diffuses to other sites to further assist such processes. The improvement in optical and mechanical properties is attributed to the increased structural order induced by the addition of hydrogen, which promotes the restructuring of weak Si–Si bonds. A reduction in dangling bond density is a consequence of this process, but the reduced dangling bond density does not correlate with, and is not directly responsible for, the reduced sub-gap (region A) absorption nor the mechanical loss. Other defects, such as those known to lead to sub-gap absorption in crystalline silicon, are necessary to explain the results.

Hydrogen reduces the optical absorption in MS *a*-Si films by orders of magnitude, reaching 9.0, 0.42, and 0.13 cm^{−1}

at 1064, 1550, and 2000 nm, respectively. Previous results on *a*-Si:H grown by CVD reported $\alpha = 1 \text{ cm}^{-1}$ at 1064 nm [38], suggesting that further reductions in optical absorption at this wavelength may be achieved in films grown by PVD. Compared to *a*-Si films with the lowest reported optical absorption [30,31], prepared by IBS and annealed in vacuum up to 500 °C, MS *a*-Si:H also shows a significant reduction in the absorption coefficient by more than 45 times at 1064 nm, 170 times at 1550 nm, and 15 times at 2000 nm. Hydrogen also reduces mechanical loss at room temperature, and hence thermal noise, while only slightly reducing the refractive index. These results show that *a*-Si:H is a strong candidate as the high-index material for the development of mirror coatings in gravitational-wave detectors, both in multimaterial coatings for room-temperature operation at 1064 nm and in traditional coatings for cryogenic-temperature operation at 1550 nm or 2000 nm.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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